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JSContact Profile for RESTful Provisioning Protocol (RPP)

Abstract

This document defines a JSContact usage profile, conforming to the rules described in [RFC10050], for the RESTful Provisioning Protocol (RPP). The JSContact Profile for RPP specifies how the JSContact format defined in [RFC9982] can be used as a standardized representation of contact information within RPP operations, enabling interoperability and consistency across RPP implementations that manage contact data.

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1. Introduction

JSContact [RFC9982] is a flexible, JSON-based format for representing contact information. Its flexibility allows the same data to be expressed in multiple valid ways, which can hinder interoperability in a provisioning context where data must be stored and interpreted consistently.

This document defines a JSContact usage profile for the RESTful Provisioning Protocol (RPP), restricting the set of JSContact constructs that RPP clients and servers are required to support when provisioning contact information.

1.1. Relationship to the RDAP JSContact Profile

The Registration Data Access Protocol (RDAP) [RFC9083] provides read-only access to registration data managed by RPP. Both protocols use JSContact for contact representation, but with profiles suited to their respective roles.

The RDAP JSContact profile [I-D.ietf-regext-rdap-jscontact] defines the constructs used when serving registration data to read-only consumers. The RPP profile, defined in this document, is a subset of these constructs. All JSContact constructs permitted by the RPP profile can be used for RDAP.

The converse does not hold. RDAP supports additional JSContact constructs that are useful in the read-only context of RDAP responses but that may hinder interoperability. These constructs are not present in RPP requests and responses.

2. Terminology

In this document, the following terminology is used.

RESTful Provisioning Protocol or RPP - The protocol described in this document.

URL - A Uniform Resource Locator as defined in [RFC3986].

RPP client - An HTTP user agent performing an RPP request.

RPP server - An HTTP server responsible for processing requests and returning results in any supported media type.

3. Conventions Used in This Document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT","SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

4. Using JSContact in RPP

4.1. JSContact Profile for RPP

Since JSContact is a generalized representation of contact data, many of its capabilities are not optimized for interoperable use in RPP payloads. This document therefore defines a usage profile for JSContact that simplifies implementation for both RPP clients and RPP servers.

The JSContact profile for RPP is compliant with the rules described in [RFC10050]. The profile properties are formally listed in Section 4.1.12. All types mentioned in this section refer to the JSContact specification [RFC9982].

4.1.1. Version

The Card `"version"` member value MUST be `"2.0"`.

4.1.2. Kind

If present, the Card `"kind"` member value MUST be `"individual"` (default) or `"org"` to represent an individual or an organization, respectively.

4.1.3. Language

The Card `"language"` member SHOULD be set when localizations are specified (i.e., when the `"localizations"` member is present and non-empty).

4.1.4. Name

Other than the optional `"@type"` member, the Name type MUST include only the `"full"` member.

4.1.5. Organizations

Other than the optional `"@type"` member, the Organization type MUST include only the `"name"` member.

4.1.6. Addresses

Other than the optional `"@type"` member, the Address type MUST include only the `"components"` and `"countryCode"` members.

Other than the optional `"@type"` member, the AddressComponent type MUST include only the `"kind"` and `"value"` members. The `"kind"` member value MUST be one of `"name"`, `"locality"`, `"region"`, or `"postcode"`.

When both an internationalized and a localized version of an address are used, the localized version MUST be placed in the `"localizations"` map keyed by an appropriate [BCP47] language tag.

4.1.7. Emails

Other than the optional `"@type"` member, the EmailAddress type MUST include only the `"address"` member.

4.1.8. Phones

The Phone type MUST include the `"number"` member and MAY include the `"features"` member. If present, the `"features"` member MUST contain only the `"voice"` or `"fax"` keys, and the value associated with each key MUST be `true`. When the `"features"` member is absent, the phone number is assumed to be a voice number.

4.1.9. Links

The Link type MUST include the "uri" member and MAY include the "kind" member. When the "kind" member is present, its value MUST be "contact".

4.1.10. Map Keys

A JSContact map key MUST comply with the Id type definition in [RFC9553]. RPP implementations MUST use the following predefined keys:

- "org" in the "organizations" map for a single preferred organization.
- "addr" in the "addresses" map for a single preferred postal address.
- "email" in the "emails" map for the preferred email address.
- "voice" in the "phones" map for the preferred voice number.
- "fax" in the "phones" map for the preferred fax number.
- "url" in the "links" map for a preferred contact URL; the "kind" member of the Link object MUST NOT be set.
- "contact-uri" in the "links" map for a contact URI as defined in [RFC8605]; the "kind" member of the Link object MUST be set to "contact".

When internationalized and localized versions of an organization, postal address, or email exist, the internationalized version MUST be included under the predefined map key, while the localized version MUST be placed in the "localizations" map as described in Section 4.1.11.

4.1.11. Localizations

If present, localized variants of name, organization, postal address, and email MUST be added to the "localizations" map. RPP implementations MUST expand all localizations, meaning a nested PatchObject key of the form "{key1}/{key2}/.../{keyN}" MUST NOT be used.

The number of provided localizations MUST NOT exceed the number of languages supported by the RPP server. The language of each localization MUST be indicated by a [BCP47] language tag as the key in the "localizations" map.

The following example of a Card includes a localized version of the postal address:

```
{
  "@type": "Card",
  "version": "2.0",
  "language": "en",
  "name": {
    "full": "Joe User"
  },
  "organizations": {
    "org": { "name": "Example Org" }
  },
  "addresses": {
    "addr": {
      "components": [
        { "kind": "name",     "value": "Main Street 1" },
        { "kind": "locality", "value": "Ludwigshafen am Rhein" },
        { "kind": "region",   "value": "Rhineland-Palatinate" },
        { "kind": "postcode", "value": "67067" }
      ],
      "countryCode": "DE"
    }
  },
  "localizations": {
    "de": {
      "addresses": {
        "addr": {
          "components": [
            { "kind": "name",     "value": "Hauptstraße 1" },
            { "kind": "locality", "value": "Ludwigshafen am Rhein" },
            { "kind": "region",   "value": "Rheinland-Pfalz" },
            { "kind": "postcode", "value": "67067" }
          ],
          "countryCode": "DE"
        }
      }
    }
  }
}
```

Figure 1: Example of handling localizations in JSContact

4.1.12. Profile Properties

The following entries define the properties of the JSContact profile for RPP registered in the JSContact Profile registry as described in Section 5.1. The field names are those defined in [RFC10050]. Entry fields with empty values are omitted:

- Property Name: language
  Property Context: Card
- Property Name: kind
  Property Context: Card
  Restricted Enum Values: individual, org
- Property Name: name
  Property Context: Card

- Property Name: full
  Property Context: Name
  Restricted Attributes: mandatory
- Property Name: organizations
  Property Context: Card
- Property Name: name
  Property Context: Organization
  Restricted Attributes: mandatory
- Property Name: addresses
  Property Context: Card
- Property Name: countryCode
  Property Context: Address
- Property Name: components
  Property Context: Address
- Property Name: kind
  Property Context: AddressComponent
  Restricted Enum Values: name, locality, region, postcode
- Property Name: value
  Property Context: AddressComponent
- Property Name: emails
  Property Context: Card
- Property Name: address
  Property Context: EmailAddress
- Property Name: phones
  Property Context: Card
- Property Name: features
  Property Context: Phone
  Restricted Enum Values: voice, fax
- Property Name: number
  Property Context: Phone
- Property Name: links
  Property Context: Card
- Property Name: kind
  Property Context: Link
  Restricted Enum Values: contact
- Property Name: uri
  Property Context: Link
- Property Name: localizations
  Property Context: Card
  Restricted PatchObject: yes

The `"@type"` member is permitted on every JSContact object in this profile and is therefore not listed above.

Client validation MUST return an error if any JSContact properties other than those listed above are present.

5. IANA Considerations

5.1. JSContact Profile Registry

IANA is requested to register the following entry in the "JSContact Profile" registry [RFC10050]:

Name: rpp
Version: 1
Reference: Section 4.1.12 of this document

TODO

6. Internationalization Considerations

TODO

7. Security Considerations

RPP servers should impose appropriate limits on the size and complexity of JSContact objects received from RPP clients, including the number and size of expanded localizations.

TODO

8. Change History

8.1. Version 00 to 01

- Fixed duplicate text (Issue #14)
- Aligned with draft-ietf-regext-rdap-jscontact-27
- Allowed the Address `"countryCode"` member and removed Address `"full"` from the profile properties (Issue #19)

9. References

9.1. Normative References

[BCP47] Best Current Practice 47, <https://www.rfc-editor.org/info/bcp47>.
At the time of writing, this BCP comprises the following:

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[RFC8605] Hollenbeck, S. and R. Carney, "vCard Format Extensions: ICANN Extensions for the Registration Data Access Protocol (RDAP)", RFC 8605, DOI 10.17487/RFC8605, May 2019, <https://www.rfc-editor.org/info/rfc8605>.

Acknowledgements

TODO

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